

Sarah Graup

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Technical Skills

- **Languages:** Javascript, Python, Java, C++, Typescript, SQL, PHP
- **FrontEnd:** HTML, CSS, JQuery, Bootstrap, React, React Native, React Testing Library, JQuery, Jest, Tanstack, MUI
- **BackEnd:** Flask, Node.js, Express, Django, MySQL, PostgreSQL, SQLAlchemy, Unittest

Software Engineer Experience

Software Engineer Intern

Super{set} - Codei.ai

May 2024 - August 2024

San Francisco, CA

- Designed and led the development of a Chrome extension that automates job application saving through web scraping and HTTP API requests, streamlining the process directly from job boards.
- Built a web application using React, Typescript, and MUI to manage job applications, generate practice questions, and evaluate code repositories.
- Integrated Stytych for secure user authentication, ensuring seamless integration between the extension and web app, and implemented Rollbar for robust error tracking.
- Developed a Python and FastAPI backend to handle data storage and processing, utilizing Github API to access user repositories and OpenAI API for customized question generation and advanced code evaluation.
- Led user feedback sessions to refine features and improve user experience.

Software Engineer Intern

School Information Systems Rithm School

April 2023 - June 2023

San Francisco, CA

- Developed a dynamic Django-built web application to refactor and implement functionality for students and administration in an 80k LOC learning management system. Achieved 98% test coverage by designing and executing unit and integration tests with Django's testing framework and Factory Boy.
- Architected and built announcement notifications in Django with Django Rest Framework and Django Template Language; created a streamlined admin panel, improving navigational simplicity through the use of Django Forms and Factory LazyAttributes for dynamic content generation.
- Automated student pairing of weekly projects employing Irvings Stable Roommates Algorithm. Addressed and resolved identified corner cases through the development of specialized solutions.

Projects

SUDOKU SOLVER

[Live Demo](#) | [Github](#)

- Engineered a Sudoku Solver leveraging the Conflict-Driven Clause Learning (CDCL) algorithm in Node.js and Express, optimizing puzzle resolution through advanced algorithms featuring recursion for logic handling and asynchronous puzzle loading.
- Utilized Unit Propagation, MRV, and Degree Heuristics to manage solver states, enabling efficient backtracking and conflict resolution to enhance solving accuracy and speed.
- Developed an interactive React frontend, utilizing Material-UI, enabling interactive puzzle selection, real-time solving visualization through Axios NPM, and comprehensive user control over the solving process

PIX.LY

[Github](#)

- Engineered a feature-rich image editor and display application combining React, Flask, and Amazon S3, optimizing for scalability and secure data storage. Incorporated Axios for efficient API requests and achieved 100% test coverage with Jest, demonstrating thorough functionality validation and reliability.
- Implemented user-friendly photo editing features using the PIL library and enhances search functionality with AWS-hosted Exif data, enabling advanced image searches by description, location, and camera make/model.
- Crafted an interactive photo gallery utilizing Reactstrap for intuitive UI design, React Router for seamless navigation, and React Context for state management across components.

Education

Rithm School | San Francisco, CA | 2023

Full Stack Web Development Bootcamp

2023

San Francisco, CA

UCLA | Los Angeles, CA | 2018-2022

B.S. Cognitive Science, Computer Programming Specialization

2022

Los Angeles, CA